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$$\sum_{n=1}^{\infty} \frac{n+1}{n^3+3n^2} > \sum_{n=1}^{\infty} \frac{1}{n^2}$$

$$\lim_{n\to\infty} \frac{\frac{n+1}{n^3+3n^2}}{\frac{1}{n^2}} = \lim_{n\to\infty} \frac{n^2(n+1)}{n^3+3n^2} = 1$$

Verhoudingscriterium: $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent, dus $\sum_{n=1}^{\infty} \frac{n+1}{n^3+3n^2}$ is convergent.

References